



Guseyn Ismayylov

SENIOR FULL-STACK SOFTWARE ENGINEER · TECHNICAL LEADER

LEGAL NAME: GUSEIN ISMAIYLOV · REMOTE · GEORGIA 🇧🇪

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Senior Full-Stack AI Software Engineer and **Technical Leader** with **10 years** building production platforms end to end — backend, frontend, infra, and data. Specialized in **agentic AI**, LLM orchestration, RAG pipelines, and intelligent middleware in real products, not demos. Equally comfortable with **Node.js, Java, TypeScript, React/Next.js, PostgreSQL, Redis, AWS, and Docker**. Background in Applied Mathematics; author of [Mathematics of Neural Networks](#), [Unison](#), [nodes.js](#), and [EHTML](#). Daily builder with **Cursor** (multi-model Composer), **Google Gemini**.

Skills

AI Agents & LLM Engineering — Agent orchestration, multi-agent workflows, LLM middleware, RAG pipelines, vector embeddings & semantic search, MCP/tool-calling, function calling, prompt engineering & tool chains, streaming inference, eval suites & guardrails, AI observability, workflow automation (n8n), intelligent product features in production.

AI-Native Development — Cursor IDE (Composer — Claude Opus/Sonnet, GPT, Gemini, Grok model routing), Google Gemini (API & workspace) — AI-accelerated full-stack delivery, autonomous refactors, test generation, and architecture iteration at scale.

Generative AI Stack — OpenAI API, Anthropic Claude, Google Gemini, embeddings & reranking, structured outputs, context-window optimization, hybrid retrieval (PostgreSQL/pgvector, Redis, S3), token & cost tuning for production AI workloads.

System Architecture — Distributed Systems, Microservices, Event-Driven Architecture (EDA), Multi-tenant SaaS, High-Load MVPs, System Design, Scalability, High Availability & Reliability Engineering.

Backend Ecosystem — Node.js (Express, Fastify, NestJS), JavaScript/TypeScript, Java (Spring Boot), PHP, Python (FastAPI), NX Monorepos, RESTful APIs, GraphQL, Serverless Architecture.

High-Performance UI/UX — Web Components, EHTML (Founder), React, Next.js, React Server Components (RSC), React Context, React Query, Tailwind CSS, Material UI, Zustand, Angular, Real-time SVG/Canvas Rendering, Shadow DOM, Micro-frontends, HTML5/CSS3, Vite, Esbuild.

Data Engineering & Storage — PostgreSQL, MySQL, MariaDB, Redis (Caching, Pub/Sub), ClickHouse (OLAP), Snowflake, Supabase, Drizzle ORM, DynamoDB, Prisma, Apache Kafka, RabbitMQ, MongoDB, Elasticsearch, Database Optimization.

Cloud Platforms & DevOps (IaC) — AWS (EC2, ECS, S3, RDS, Lambda, DynamoDB, CloudWatch), GCP (Cloud Run, App Engine, BigQuery, VPC), Azure, DigitalOcean, Docker, Kubernetes (K8s), Terraform, GitHub Actions, GitLab CI/CD, Infrastructure as Code (IaC).

Observability & Telemetry — OpenTelemetry, Prometheus, Grafana, New Relic, Graylog, Datadog, AWS CloudWatch, ELK/Elastic Stack.

Protocols & Integrations — WebRTC (Xirsys), WebSockets, Stripe API (Payment Infrastructure), Postmark (Email Delivery), Azure AD (OAuth2/OIDC), Secure Identity Management.

Testing & Quality Assurance — LLM eval suites, golden-set regression, prompt regression testing, RAG retrieval quality checks, guardrail & safety testing, Testcontainers, Jest/Vitest, Playwright, Supertest, Postman/Newman, TDD/BDD, coverage & reporting (Allure, Istanbul), CI test gates, AI-assisted test generation.

AI — Cursor (Composer — Claude, GPT, Gemini, Grok models), Google Gemini, n8n; LLM-assisted coding, review, and refactoring.

Work Experience

InstruxMusic — Senior Full-Stack Developer, Tech Lead

NOV 2024 – PRESENT · REMOTE · USA

- Architected and accelerated the launch of a complex ed-tech ecosystem, delivering a production-ready MVP in under 3 months using a streamlined stack of PostgreSQL, Docker, and DigitalOcean.
- Engineered a scalable multi-tenant scheduling engine and automated invoicing system, integrating Stripe Payments to establish seamless transactional workflows and eliminate manual billing overhead.
- Developed a proprietary, low-latency video conferencing solution from scratch using vanilla Web Components and Xirsys (STUN/TURN), bypassing heavy third-party framework dependencies to achieve highly efficient real-time streaming, chat, and data channel communication.
- Designed high-deliverability notification pipelines by implementing Postmark, significantly cutting down user communication friction through automated transactional workflows.
- Drive technical strategy and product vision in alignment with business scaling and pricing frameworks, managing execution speed while maintaining a high-performance, low-bloat architecture.

[View Recommendations](#)

Congrifly — Full Stack Senior Software Engineer, Tech Lead

OCT 2023 – OCT 2024 · REMOTE · GERMANY/ITALY

- Directed technical strategy and system observability by architecting a specialized monitoring system using Grafana and AWS CloudWatch, enabling real-time detection of payment transaction anomalies and resolving critical infrastructure bottlenecks.
- Reduced AWS Lambda operational costs by 20% through targeted workflow optimizations, memory tuning, and improved data-processing logic across serverless environments.
- Spearheaded the modernization of legacy microservices, dramatically improving concurrency patterns, query execution, and high-load stability by migrating complex datasets to ClickHouse (OLAP) and Snowflake.
- Enhanced large-scale data ingestion and asynchronous messaging pipelines by leveraging AWS SQS and robust Pub/Sub architectures, ensuring zero-loss system reliability and high availability under heavy transactional surges.
- Engineered a robust AI middleware layer alongside low-latency data tiers utilizing DynamoDB and S3 to facilitate real-time streaming data processing and intelligent user analytical insights.
- Led an end-to-end frontend redesign and streamlined communication systems by integrating AWS SES for transactional mailings and optimizing the React ecosystem with advanced caching and sophisticated state management.

FxPro — Senior Software Engineer

NOV 2022 – OCT 2023 · LIMASSOL, CYPRUS

- Orchestrated infrastructure for 60+ microservices, delivering a 20% improvement in system reliability and a 15% reduction in operational downtime.
- Created a centralized Authentication Service with Azure Active Directory integration and successfully migrated payment configurations for 20+ global systems across 50+ countries.
- Optimized data consistency and performance across distributed services by leveraging Redis for caching and PostgreSQL for mission-critical financial data.
- Streamlined CI/CD pipelines using Artifactory in GitLab, reducing deployment cycles by 25% and ensuring high-availability across global environments.
- Implemented full-stack observability across 25+ services using New Relic, Graylog, and Grafana, enabling real-time bottleneck identification and rapid incident response.

[View Recommendations](#)

Cognitran — Senior Software Engineer

MAR 2021 – OCT 2022 · KRAKOW, POLAND

- Developed mission-critical, offline-first automotive diagnostic software and automated technical documentation pipelines for enterprise clients including Ford and Jaguar.

Huawei — Senior Software Engineer

JAN 2020 – MAR 2021 · MOSCOW, RUSSIA

- Designed high-performance cloud infrastructure components, optimized internal developer tooling, and crafted high-throughput custom data parsers.

US Mobile — Senior Software Engineer

FEB 2019 – JAN 2020 · REMOTE · USA

- Helped design technical solutions and created a roadmap for migrating infrastructure to Google Cloud Platform (GCP).
- Successfully oversaw the implementation of this replatforming, ensuring a smooth and efficient transition.
- Spearheaded the full automation of report processing from third-party sources, utilizing GCP services to significantly improve data processing and operational efficiency.

Grid Dynamics — Intern to Senior Software Engineer

DEC 2015 – JAN 2019

- Progressed from Intern to Senior Engineer, delivering high-impact cloud-native R&D internal tooling for major enterprise clients including Google, Raymond James, and Home Depot.

Certifications

Oracle Certified Associate Java SE 8
Programmer

Open Source

EHTML, Nodes.js, e-ui, Mathematics
of Neural Networks

Education

Applied Mathematics and Computer
Science — *Saratov State University*,
2012 - 2016
[Bachelor's Thesis \(PDF\)](#)